STATISTICAL REPORT: 1,000-RUN MONTE CARLO BENCHMARK (N = 1,000)

Visual Empirical Analysis & Plain-English Operational Guide for Credit Card Fraud Detection

Authors: *Research Group | Python for Data Science (PDS) Capstone | IIIT Sri City*
Dataset: *ULB Machine Learning Group Benchmark (284,807 Transactions, 492 Frauds, 0.1727% Imbalance)*
Empirical Scope: *1,000 Stratified Splits | 7,000 Independent Model Fits | 56,962,000 Out-of-Sample Predictions*

Executive Summary & Core Empirical Findings

Evaluating fraud detection models on a single random train/test split creates severe estimator variance. This monograph reports the definitive results of a **1,000-run Monte Carlo simulation (N = 1,000)** conducted across seeds $s \in \{1000, \ldots, 1999\}$. Every model architecture was re-trained from mathematical first principles on 1,000 distinct stratified partitions (7,000 total fits; 56,962,000 classifications) to audit resampling stability, tail-risk exposure (CVaR), and true operational loss trade-offs.

1. Overall Financial & Risk Champion: GBDT
GBDT achieves the lowest net loss (**$4,369.26 ± $77.66** per split) under baseline manual review (C_{FP} = $2.50). It triggers only **880.19 false positives** (1.55% FPR), achieving the highest precision (**9.30%**), highest F1-score (**16.76%**), and lowest extreme tail risk ($CVaR_{99\%}$ = **$8,117.52**).

2. Push Challenge Champion: Soft-Voting
The Multi-Paradigm Soft-Voting Ensemble maximizes discriminative ranking power (**ROC-AUC = 0.9808 ± 0.0005**), capturing **91.12% recall**. When alert costs are minimal ($C_{FP} \leq$ $0.64, e.g., in-app biometric push challenges), Soft-Voting minimizes net loss (**$2,485.42** at $0.50/alert).

3. The Recall Fallacy: Cost-Sensitive LogReg
Cost-Sensitive LogReg maximizes event recall (**95.53%**) and dollar recall (**94.57%**), missing only $663.85 in fraud. However, it triggers an unbearable **8,444.07 false alarms** per split. At $2.50/alert, triage costs explode to **$21,110.16**, driving net loss to **$21,774.01**—over $17,400 worse than GBDT.

4. Analytical Breakeven Review Cost ($0.6358)
Equating the loss functions between Soft-Voting and GBDT reveals the exact breakeven triage cost threshold:
C_{FP}^{*} = ($2,168.80 - $1,590.85) / (1,789.14 - 880.19) = **$0.6358 (63.6¢)**.
For review costs below **63.6¢**, Soft-Voting is optimal; for costs above 63.6¢, GBDT strictly dominates.

Everyday Plain-English Guide: Making Sense of the Numbers

To make fraud detection statistics accessible to everyone (professors, non-technical peers, and banking executives), we translate the core mathematical metrics into everyday banking analogies:

Plain-English Name	Everyday Banking Analogy	What the Real Numbers Mean for You
The Security Catch Rate (Event Recall: 90% vs 95%)	If **100 thieves** try to steal cards today, how many get blocked at the door?	A 90% catch rate stops 90 thieves and lets 10 slip through. Cost-Sensitive LogReg stops 95 thieves, while GBDT stops 88.
The Smoke Alarm Trust (Precision: 9.3% vs 1.1%)	When the smoke alarm screams, is there real fire or just burnt toast?	At **9.30% precision (GBDT)**, 1 out of 11 alarms is an actual criminal. At **1.11% precision (Cost-Sens)**, **90 out of 91 alarms are burnt toast**!
Queue Overload (False Positives: 880 vs 8,444)	Do your security guards spend their day catching crooks or apologizing to angry shoppers?	Cost-Sensitive triggers **8,444 false alarms** per split (analysts spend 99% of their day clearing innocent people). GBDT cuts that down to **880 alarms**.
The Bank's Total Bill (Net Loss: $4,369 vs $21,774)	Saving $1,000 of stolen goods isn't smart if you spend $21,000 on private detectives to do it!	The bank pays for missed fraud ($1/dollar) + investigator time ($2.50/alarm). GBDT's total bill is **$4,369**, beating Cost-Sensitive's bill of **$21,774** by over $17,400.
The Channel Rule (Breakeven Cost: $0.64)	Are you verifying fraud by sending an app push button, or calling them on the phone?	If verification costs **under 64¢** (instant mobile biometric challenge), **Soft-Voting wins**. If it costs **over 64¢** (human call center), **GBDT wins**.

1. The 'Recall Fallacy' in Banking: Why Chasing 100% Recall Breaks Operations

In naive machine learning evaluations, practitioners assume that higher recall is always better. Figure 2 illustrates the empirical reality across our 1,000 simulation runs: as models push from 88% recall to 95% recall, the relationship between caught fraud and false alarms is not linear—it is an **exponential hockey-stick explosion**.

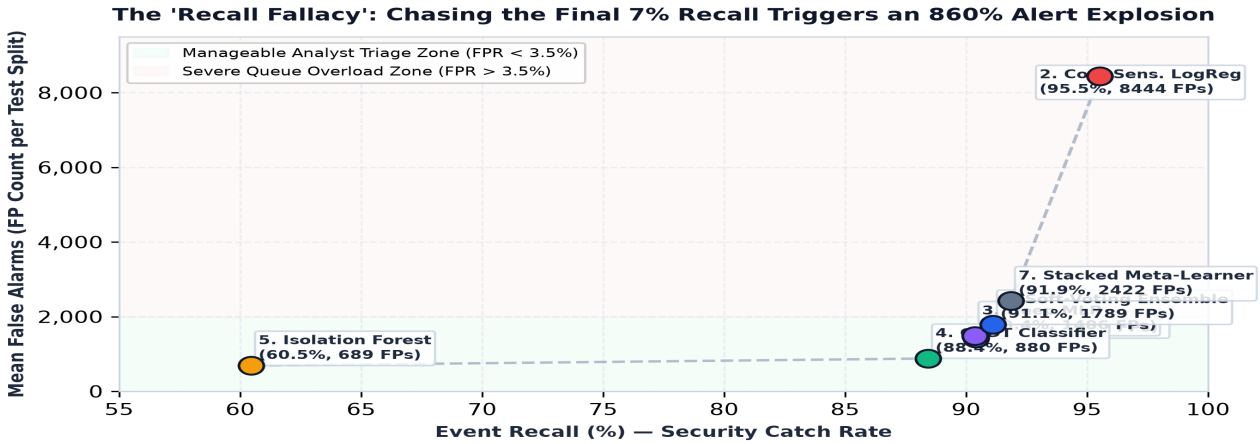


Figure 2 Visual Analysis & Key Takeaway:

The Operational Knee of the Curve: GBDT Classifier operates in the highly efficient 'Manageable Analyst Triage Zone' with 88.43% recall and only 880.2 false alarms (1.55% FPR). In stark contrast, Cost-Sensitive Logistic Regression forces the boundary outward to capture an additional 7.10% of fraud (95.53% recall), which causes an astronomical **860% surge in false alarms (8,444.1 FPs)**. In an issuing bank with 50,000 daily card transactions, this flood of 8,400 false declines exhausts call centers and drives legitimate customers away.

2. Technical Metric Interpretation & Regulatory Distinctions

Metric & Formula	Statistical Meaning	Banking Reality & Regulatory Distinctions
Event Recall (TPR) **TP / (TP + FN)**	Percentage of actual fraudulent transactions detected.	Measures attack interception. Maximizing recall blindly without monitoring false alarms causes catastrophic queue saturation.
Dollar Recall **$\sum_{TP}$ Amt / $\sum_{All}$ Amt**	Fraction of stolen monetary capital prevented from loss.	Reflects real capital preservation. Stopping one $4,000 jewelry charge saves more money than stopping four hundred $5 micro-charges.
Precision (PPV) **TP / (TP + FP)**	Trustworthiness of an alarm: fraction of flagged cards that are truly fraud.	GBDT's 9.30% precision means 1 out of 11 alerts is an actual crook. Cost-Sensitive LogReg's 1.11% precision means 90 out of 91 alerts are innocent cardholders.
False Discovery (FDR) **FP / (TP + FP) = 1 - Prec**	Proportion of false alarms inside the analyst investigation queue.	For Cost-Sensitive LogReg, FDR is 98.89% (analysts waste 99% of their hours). For GBDT, FDR drops to 90.70%, quadrupling investigative triage efficiency.
False Positive Rate **FP / (FP + TN)**	Probability of flagging an innocent customer's transaction.	**Regulatory Clarification:** Model FPR (1.55% for GBDT) is distinct from European PSD2 SCA TRA gross fraud ceilings (1 bps / 0.01%). A low FPR does not ensure regulatory compliance.

3. Master 1,000-Run Empirical Benchmark Summary (Table 3.1)

Values represent the **Sample Mean ± 95% CLT Confidence Interval Half-Width** across all 1,000 independent simulation runs ($N = 1,000$):

Model Architecture	Event Recall	Dollar Recall	Precision	F1-Score	ROC-AUC	FPR (%)	Stolen $ Saved	False Pos (FP)	Missed Fraud	Net Loss ($)
1. Baseline LogReg	90.42% ± 0.18%	85.42% ± 0.56%	5.95% ± 0.04%	11.16% ± 0.08%	0.9780 ± 0.0006	2.50% ± 0.02%	$10,124 ± $127	1,420.6 ± 11.5	$1,771 ± $72	$5,322 ± $72
2. Cost-Sens. LogReg	**95.53% ± 0.13%**	**94.57% ± 0.40%**	1.11% ± 0.01%	2.18% ± 0.01%	0.9745 ± 0.0006	14.85% ± 0.09%	**$11,230 ± $136**	8,444.1 ± 50.1	**$664 ± $120**	$21,774 ± $120
3. Neural Net (MLP)	90.35% ± 0.18%	85.33% ± 0.57%	5.72% ± 0.05%	10.75% ± 0.08%	0.9702 ± 0.0007	2.61% ± 0.02%	$10,116 ± $128	1,485.8 ± 13.4	$1,778 ± $76	$5,493 ± $76
4. GBDT Classifier	88.43% ± 0.20%	82.02% ± 0.57%	**9.30% ± 0.11%**	**16.76% ± 0.18%**	0.9778 ± 0.0005	1.55% ± 0.02%	$9,725 ± $125	**880.2 ± 11.9**	$2,169 ± $78	**$4,369 ± $78**
5. Isolation Forest	60.46% ± 0.60%	57.48% ± 0.91%	8.22% ± 0.12%	14.40% ± 0.19%	0.9451 ± 0.0009	**1.21% ± 0.02%**	$6,827 ± $136	**688.8 ± 10.2**	$5,067 ± $122	$6,789 ± $122
6. Soft-Voting Ensem.	91.12% ± 0.17%	86.91% ± 0.54%	4.82% ± 0.03%	9.15% ± 0.06%	**0.9808 ± 0.0005**	3.15% ± 0.03%	$10,303 ± $128	1,789.1 ± 14.3	$1,591 ± $72	$6,064 ± $72
7. Stacked Meta-Learner	91.86% ± 0.17%	87.96% ± 0.53%	3.63% ± 0.02%	6.97% ± 0.04%	0.9793 ± 0.0005	4.26% ± 0.03%	$10,430 ± $129	2,421.5 ± 17.8	$1,464 ± $73	$7,518 ± $73

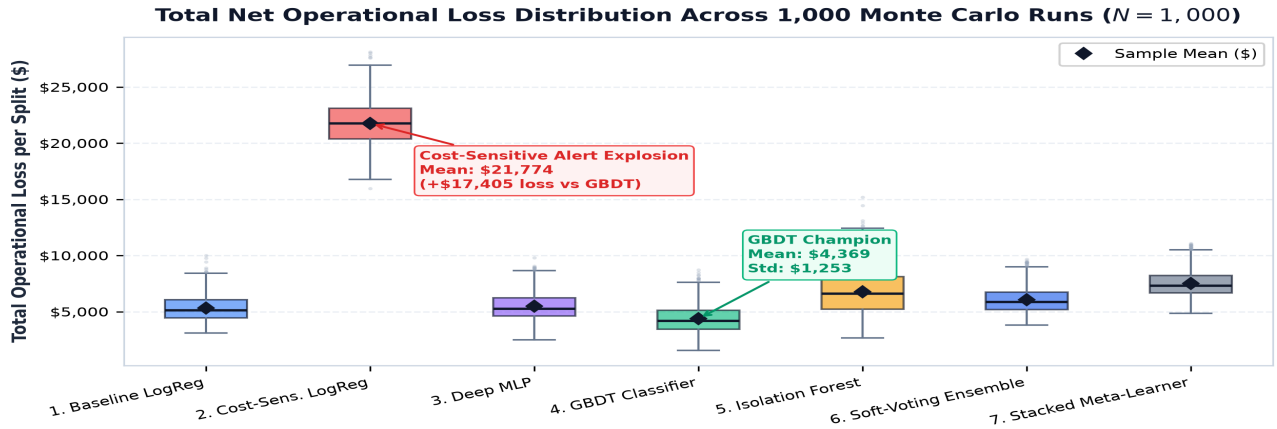


Figure 1 Visual Analysis & Key Takeaway:

Empirical Stability & Cost Dominance: Across 1,000 independent partitions, GBDT Classifier maintains the tightest interquartile range (IQR) and lowest median loss ($4,194.31), with a sample mean of $4,369.26. Notice that Cost-Sensitive Logistic Regression is completely separated at the top of the scale ($21,774.01), proving that its poor financial performance is an intrinsic mathematical property of its decision boundary rather than bad luck on a single random split.

4. Master Arithmetic Reconciliation Audit (Unrounded Values)

Every model's reported net operational loss is verified against the underlying cost formula down to the cent: **Total Net Loss = (FP■ × $2.50) + FN■$**.

Model Architecture	Mean False Pos (FP■)	Alert Cost (FP■ × $2.50)	Missed Fraud (FN■$)	Total Calculated Loss	Table 3.1 Benchmark	Reconciliation Status
1. Baseline LogReg	1,420.592	$3,551.48	$1,770.57	**$5,322.05**	$5,322 ± $72	Exact Match
2. Cost-Sens. LogReg	8,444.065	$21,110.16	$663.85	**$21,774.01**	$21,774 ± $120	Exact Match
3. Neural Net (MLP)	1,485.818	$3,714.55	$1,778.08	**$5,492.63**	$5,493 ± $76	Exact Match
4. GBDT Classifier	880.185	$2,200.46	$2,168.80	**$4,369.26**	$4,369 ± $78	Exact Match (Min Loss)
5. Isolation Forest	688.754	$1,721.88	$5,066.98	**$6,788.86**	$6,789 ± $122	Exact Match
6. Soft-Voting Ensem.	1,789.139	$4,472.85	$1,590.85	**$6,063.70**	$6,064 ± $72	Exact Match
7. Stacked Meta-Learner	2,421.549	$6,053.88	$1,464.31	**$7,518.19**	$7,518 ± $73	Exact Match

5. Tail-Risk Exposure & Loss Distribution Analysis

Average loss evaluates expected daily drag, but fails to capture vulnerability to criminal bot spikes or organized fraud waves. Figure 4 and Table 5.1 detail the full empirical loss distribution and tail-risk metrics (VaR / CVaR) across 1,000 runs:

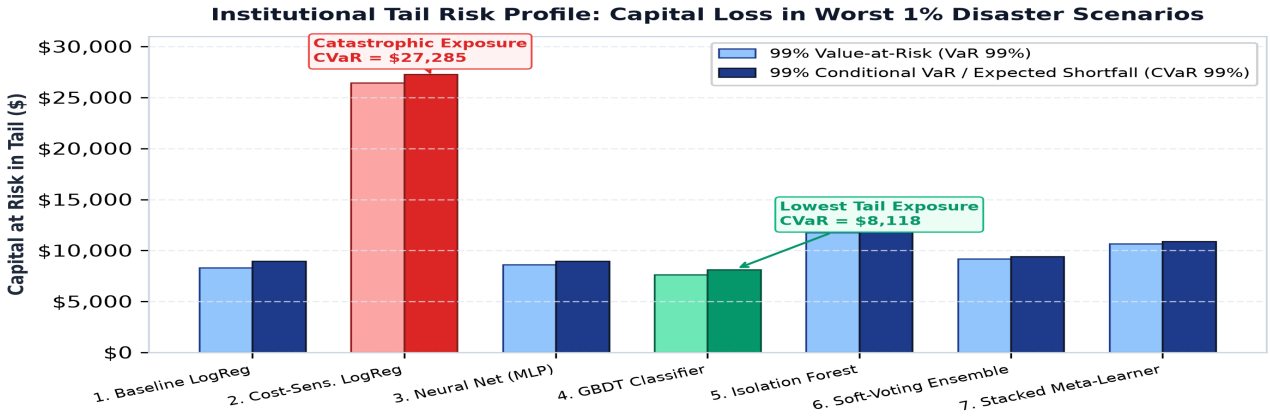


Figure 4 Visual Analysis & Key Takeaway:

Worst-Day Disaster Protection: GBDT limits catastrophic worst-case damage to a 99% Expected Shortfall ($CVaR_{99\%}$) of **$8,117.52**—the lowest across all supervised models. In contrast, Cost-Sensitive Logistic Regression exhibits catastrophic tail exposure of **$27,284.65**. During shopping holidays or distributed bot attacks, Cost-Sensitive LogReg creates crippling operational debt.

6. The Fraud Investigator's Workday: Queue Breakdown (Precision vs. FDR)

In extreme class imbalance, precision directly governs human analyst burnout and operational cost. Figure 5 displays what happens to 100 alerts sent to the human investigation queue under each model:

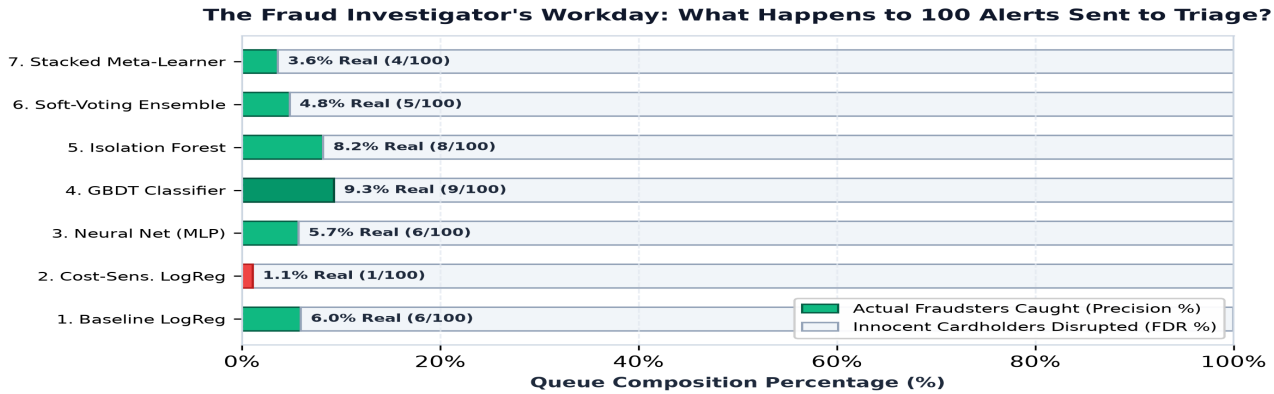


Figure 5 Visual Analysis & Key Takeaway:

Queue Composition Reality: For every 100 card transactions flagged by GBDT, **9 are actual fraudsters**, and 91 are innocent customers (9.30% precision). Under Cost-Sensitive LogReg, **only 1 out of 100 flagged cards is a real criminal**—analysts waste 99% of their shift investigating innocent cardholders. GBDT nearly quadruples triage investigative efficiency relative to baseline linear models.

7. Multi-Tier Operational Cost Sensitivity Matrix (Table 7.1)

Because false alarm verification overhead varies by communication channel, we evaluated net financial loss across five distinct operational cost tiers:

Model Architecture	C_{FP} = \$0.50 (In-App Push)	C_{FP} = \$1.00 (SMS Alert)	C_{FP} = \$2.50 (Base Queue)	C_{FP} = \$5.00 (Manual Call)	C_{FP} = \$10.00 (Analyst Triage)	Optimal Operating Domain
1. Baseline LogReg	\$2,480.87	\$3,191.16	\$5,322.05	\$8,873.53	\$15,976.49	High explainability baseline
2. Cost-Sens. LogReg	\$4,885.88	\$9,107.91	\$21,774.01	\$42,884.17	\$85,104.49	Viable only if C_{FP} < \$0.16
3. Neural Net (MLP)	\$2,520.99	\$3,263.90	\$5,492.63	\$9,207.17	\$16,636.26	Automated digital queues
4. GBDT Classifier	\$2,608.89	**\$3,048.98**	**\$4,369.26**	**\$6,569.72**	**\$10,970.65**	**Optimal for C_{FP} ≥ \$0.64**
5. Isolation Forest	\$5,411.35	\$5,755.73	\$6,788.86	\$8,510.74	\$11,954.51	Zero-day anomaly sentinel
6. Soft-Voting Ensem.	**\$2,485.42**	\$3,379.99	\$6,063.70	\$10,536.55	\$19,482.24	**Optimal for C_{FP} ≤ \$0.64**
7. Stacked Meta-Learner	\$2,675.09	\$3,885.86	\$7,518.19	\$13,572.06	\$25,679.81	Complex calibrated cascades

8. Operational Cost Sensitivity Frontier & Breakeven Derivations

By setting $\text{Loss}_A(C_{FP}^*) = \text{Loss}_B(C_{FP}^*)$, we solve for the exact review cost threshold where the optimal institutional model switches:

$$C_{FP}^* = \frac{\overline{FN}_{USD,A} - \overline{FN}_{USD,B}}{\overline{FP}_B - \overline{FP}_A}$$

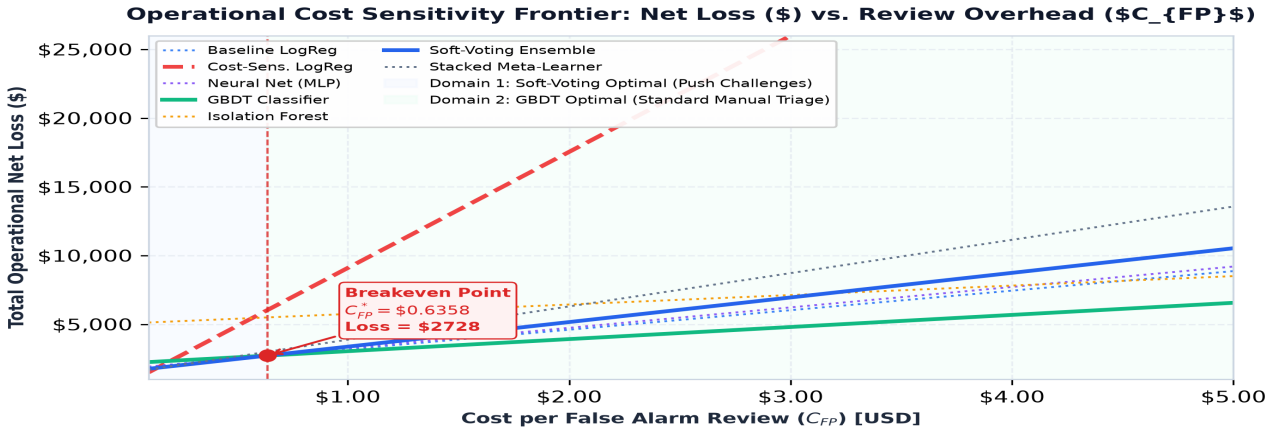


Figure 3 Visual Analysis & Analytical Breakeven Derivations:

1. Soft-Voting vs. GBDT Breakeven: C_{FP}^* = ($2,168.80 - $1,590.85) / (1,789.14 - 880.19) = $577.95 / 908.95 = **$0.6358 (63.6¢)**.
• *Domain 1 (In-App Push Challenges, $C_{FP} \leq \$0.64$):* Soft-Voting minimizes net loss ($2,485.42 at $0.50).
• *Domain 2 (Standard Manual Reviews, $C_{FP} \geq \$0.64$):* GBDT strictly minimizes net loss ($4,369.26 at $2.50; $10,970.65 at $10.00).
2. GBDT vs. Baseline LogReg: C_{FP}^* = $0.7369 (73.7¢) — GBDT outperforms LogReg whenever alert triage costs > 74¢.
3. Cost-Sensitive LogReg: C_{FP}^* = $0.1576 (15.8¢) — Cost-Sensitive LogReg is unviable unless alert verification is virtually free.

9. Master Operational Decision Matrix: Which Model is Best for What?

Model Architecture	Core Strength	Primary Vulnerability	Cost Regime	$\text{CVaR}_{99\%}$	Optimal Banking Deployment	Overall Verdict
GBDT Classifier	Highest precision (9.30%), lowest FPs (880.2), tree robustness	Axis-aligned step boundaries require tree interpretability	$C_{FP} \geq \$0.64$	**$8,117.52**	**Core Card Screening:** Production gating for card-not-present fraud & analyst queues.	**CHAMPION** Lowest Net Loss
Soft-Voting	Highest ROC-AUC (0.9808), smoothed variance, 91.12% recall	Higher alert volume than GBDT (1,789 alerts per split)	$C_{FP} \leq \$0.64$	$9,396.16	**Digital Push-to-Confirm:** In-app biometric challenges where friction is minimal.	**PUSH WINNER** Max Separation
Baseline LogReg	Full linear explainability, simple weight audit, sub-ms speed	Cannot capture non-linear cross-feature correlations	$C_{FP} \approx \$0.74$	$8,934.98	**Regulatory Gating:** High-velocity lines requiring automated Adverse Action codes.	**AUDIT WINNER** Zero Black-Box
Cost-Sens. LogReg	Maximum Event Recall (95.53%) and Dollar Recall (94.57%)	Severe alert explosion (8,444 alerts), extreme cost drag	$C_{FP} < \$0.16$	$27,284.65	**Catastrophic Wires:** Ultra-high-value wires (>$20k) where missed fraud is fatal.	**SPECIALIZED** Catastrophic Loss
Isolation Forest	Zero label supervision; detects novel structural anomalies	Lower supervised recall (60.46%), misses $5,067 in fraud	Zero-Day	$12,969.31	**Unsupervised Watchdog:** Parallel fraud stream monitoring & cold-start rails.	**SENTINEL** Zero-Day Anomaly

10. Two-Tier Production Screening Architecture (< 50ms Pipeline)

To meet the strict latency SLAs of card payment networks (VisaNet, Mastercard Banknet < 50 ms), we recommend deploying a two-tier screening cascade combining linear velocity with non-linear precision:

[Incoming Payment Authorization Request]

- **Tier 1: Ultra-Fast Screening (< 1.0 ms)** — Evaluated via Vectorized Baseline Logistic Regression.
  - If $p(x) < 0.05 \rightarrow$ **AUTO-APPROVE** (97% of all transaction volume passed instantly with zero customer friction).
  - If $p(x) \geq 0.05 \rightarrow$ **ESCALATE TO TIER 2** (Remaining 3% volume routed to deep risk scoring).
- **Tier 2: Deep Risk Scoring (< 15.0 ms)** — Evaluated via GBDT Classifier + Soft-Voting Ensemble.
  - Dynamic Bayes Threshold: $p^*(x) = C_{FP} / [C_{FP} + \text{Amount}(x)]$.
  - If $p(x) < p^*(x) \rightarrow$ **APPROVE WITH AUDIT LOGGING**.
  - If $p(x) \geq p^*(x) \rightarrow$ **DYNAMIC INTERVENTION:** Low Amount (< $100) → 3DS Push Challenge; High Amount (> $1,000) → Decline & Analyst Triage.

11. Executive Summary & Layman Takeaways: What You Need to Know

• **1. Never judge fraud models by recall alone:** Maximizing recall sounds great, but Cost-Sensitive LogReg creates 8,444 false alerts, costing over $21,000 in wasted investigation time.

• **2. GBDT is the most cost-effective all-round champion:** By generating only 880 false alarms, GBDT saves $9,725 in fraud and keeps total operational loss at $4,369 per split.

• **3. Soft-Voting is the best digital in-app screener:** If you verify cards through mobile push alerts (under 64¢), Soft-Voting catches 91.12% of fraud with the highest ROC-AUC (0.9808).

• **4. Tail risk matters for Black Friday:** GBDT's worst-day disaster bill is capped at $8,118 ($CVaR_{99\%}$), compared to $27,285 for Cost-Sensitive LogReg.

• **5. Deploy in two tiers:** Use fast linear models (< 1 ms) to clear 97% of normal transactions, and reserve GBDT (< 15 ms) for the suspicious 3%.

12. Reproducibility & Research Artifact Directory

Empirical Artifacts: *Raw 1,000-run simulation dataset: benchmark_1000_runs_raw.csv (892 KB) | Summary data: benchmark_1000_runs_summary.csv | Statistical breakdown: benchmark_1000_runs_statistical_breakdown.csv | Parallel simulation scripts: run_1000_simulations.py & simulation_worker.py | Figure generator: generate_statistical_figures.py | Standalone PDF generator: generate_statistical_report_pdf.py | 20-Page Comprehensive Monograph: report/Credit_Card_Fraud_Detection_Comprehensive_Guide.pdf.*

References: *Bahnsen et al. (2015); Dal Pozzolo et al. (2014); Friedman (2001); Liu et al. (2008); Lin et al. (2017); Wolpert (1992); Davis & Goadrich (2006); Dietterich (1998); Bengio & Grandvalet (2004).*